DOMS | Page No.
Date / /

* Case Study 1:

BrightMart is a medium sized online shopping company, customers use its website and mobile application to purchase products.
The company stores
1] Customer name
2] Email addresses
3] Phone Numbers
4] Delivery Addresses
5] Passwords
6] Order information
7] Payment related information
BrightMart has around 100 employees
The company recently hired several new employees.
To make things easier, the I.T department gave employees access to different internal systems.
However, the company has the following security weakness:

- Some employees use simple passwords such as BrightMart123
- Many employees use the same password for multiple system.
- Software updates are sometime delayed
- The company does not regularly conduct employees cyber security training
- There is no multifactor authentication for some important accounts.
- Employees can access sensetive information that they do not necessarily need for their jobs.
- Backups are performed only once a month.

DOMS | Page No.
Date / /

* The incident
One Monday morning, an employee named Rahul receives an Email.
Subject: Urgent - Your Account will be suspended.
The email appears to come from the company's IT department.
It says:
"Your company account will be suspended today. Click the link below and verify your password immediately."
Rahul clicks the link
The page looks almost identical to the company's login page
He enters his username and password
Rahul does not realise that the website is controlled by an attacker

The attacker now has Rahul's login credentials.

Stage1 Unauthorised Access
The attacker uses Rahul's credentials to access Bright Mart's internal system.
Because Rahul uses the same password for another system, the attacker is also able to access that system.
The attacker discovers that Rahul has access to customer information.
The attacker downloads a large amount of customer data

Q What cyber security problem has occured.

Stage 2: Malware attack.
The attacker sends another email to Rahul containing a malacious attachment.
Rahul opens the attachment.
Malware is installed on his computer.
The malware begins communicating with the attacker system.
The attacker now attempts to move from Rahul's computer to other system within BrightMarts network.
Question: Which vulnerability allowed the attacker to move further into the organisation?

Stage 3: Ransomware:
Two days later, employees arrive at work and discover that they cannot access important files.
A message appears on their computer
"Your files have been encrypted. Pay the ransom to recover them."
The company's customer database and several important business files are encypted.
Unfortunately, the company's most recent backup was almost a month old.
Bright Mart website stops working. Customers cannot place order. The customer support team cannot access customer records.

DOMS | Page No.
Date / /

Step 4:
Bright Mart now faces several problem.
Financial Impact - The company does loses sales because its website is unavailable
Data Impact - Customer information may have been stolen.
Operational Impact - Customer employees cannot access important systems.
Reputation Impact : Customer lose confidence in comp.
Legal Impact - The company may face regulatory or legal consequences depending on what information was exposed & which laws apply.

* Group 1 : Identify the threats :
  What threats are present in the case?
* Group 2 : Identify the vulnerabilities
  What weakness in BrightMart security allowed the attack to succeed?
* Group 3 : Apply the CIA triad
  Was confidential information exposed?
  Was information changed or damage
  Were system available to legitimate users?
* Group 4 : Security Solution :
  Who is responsible for cybersecurity?
  Was phishing the only problem?
  What was the biggest vulnerability

DOMS | Page No.
Date / /

Solution (Case Study)

1] What cybersecurity problem has occured?

Possible answers:
- Phising
- Credential theft
- Unauthorized access
- Data breach

2] Which vulnerability allowed the attackers to move further into the organisation?

- Weak passwords
- Excessive user privileges
- Lack of MFA
- Poor network security
- Lack of employee awareness
- Outdated software

3] What threats are present in this case?

- Phising
- Malware
- Ransomware
- Credential theft
- Unauthorised access
- Data theft.

DOMS | Page No.
Date / /

4] What weakness in BrightMart security allowed the attack to succeed?

Vulnerability	How it helped the attacker
Weak passwords	Easier to compromise accounts
Password reuse	One stolen password gives access to multiple system
No MFA	Stolen password may be enough to login it.
Lack of training	Employee falls for phishing
Delayed Updates	Vulnerable software may be exploited
Excessive Privileges	Attacker can access more data

Infrequent backups difficult to recover after ransomware.

5] Was confidential info exposed?

Yes [Customer information was accessed by the attacker]

Was information changed or damaged?
Yes (Potentially) [The attacker could modify or encrypt files]

Were system available to legitimate users?
No [The website and internal system became unavailable because of ransomware]

So the [CIA] was affected.

6] What should BrightMart do to prevent this from happening again?

- Use strong password
- Implement multi-factor authentication (MFA)
- Provide regular ~~to~~ cybersecurity awareness training
- Conduct phishing awareness exercise
- Keep software updated
- Follow the principle of least perivilege
- Segment important networks
- Maintain regular backups
- Encrypt sensitive data
- Deploy antivirus protection
- Monitor suspicious login activity
- Develop an incident-response activity.

DOMS | Page No.
Date / /

7] Who is responsible for cybersecurity?

Not only the I.T department. Every employee use the organisation's system has a role in cybersecurity.

8] Was Phishing the only problem
No. Phishing was the initial attack, but weakness such as password reuse, a lack of MFA excessive privellege, poor backups, & insufficient training allowed the attack to become much more damaging.

9] What was the biggest vulnerability

1] "Lack of employee awareness was a major vulnerability because the attack started when Rahul clicked the phishing link."

2] "Poor backup practices were the biggest problem because BrightHeart could not quickly recover from Ransomware."

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